Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3411-WE01

Title: **MEng/BEng:** Solid Mechanics 3		
Time Allowed:	2 hours	
Additional Material provided:	Yes - Formula Sheet included as part of exam paper	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Q1

Ten years after graduation you have set up your own fatigue consultancy and have won a contract with a commercial aircraft manufacturer to undertake an analysis of a wing flap. The part is made from 7075-T6 aluminium alloy (with mechanical properties in Table Q1). The company reports that they have had problems with edge cracks in a plate forming part of the flap assembly. The plate is designed to withstand a tensile stress that undergoes 30,000 load cycles per month with a mean stress of $\sigma_Y/6$ and with a stress ratio $R = 1/3$ when the flap is deployed.

(a) Show that the critical length of an edge crack, a_c, is 6.24 mm. State and justify any important assumptions. **[20%]**

(b) Show that the crack growth, using the Paris law, may be described by the integral statement

$$1309 \int_{a_0}^{a_1} a^{-8/5} da = \int_{N_0}^{N_1} dN$$

and briefly describe the range of the values of crack length, a, over which the formula might be reliable. **[20%]**

(c) The part is monitored every three months for development of edge cracks using equipment which your client tells you can detect cracks as small as 0.25 mm. Show that, once the minimum detectable crack length has been reached, the remaining life of the component is approximately 9 months. **[10%]**

(d) On the basis of your report, your client plans to release a memo on monitoring schedules to their maintenance crews. An extract from the draft memo reads "*We suggest maintaining our standard monitoring interval of three months. In that time, a crack that is just undetectable will grow from 0.25 mm to 2.25 mm, being one third of the way to failure and therefore giving a factor of safety of 3. The part should be replaced when the monitored crack length exceeds 2.25 mm*". They ask for your recommendations before releasing the memo. What is your response to the company? **[50%]**

Property	Notation	Value
Yield stress	σ_Y	510 MPa
Fracture toughness	K_{Ic}	20 MPa$\sqrt{\text{m}}$
Paris law coefficient	C	5.7×10^{-11}
Paris law coefficient	m	3.2

Table Q1: Mechanical properties of alloy
(Paris law coefficients assume units of MPa$\sqrt{\text{m}}$ and m/cycle)

continued

Q2

Shape functions are at the heart of finite element analysis as they provide a way to interpolate between the displacement at nodes to any point in the finite element mesh.

(a) Giving all of the steps in your derivation and clearly stating any assumptions that you make, show that the shape function for node 1 of the one dimensional cubic finite element shown in Figure Q2 is

$$N_1 = \frac{1}{16}(-1 + \xi + 9\xi^2 - 9\xi^3)$$

where $\xi \in [-1, 1]$ is the local coordinate associated with the element. Assume that the nodes are equally spaced along the element. Sketch the variation of the shape function over the element. **[50%]**

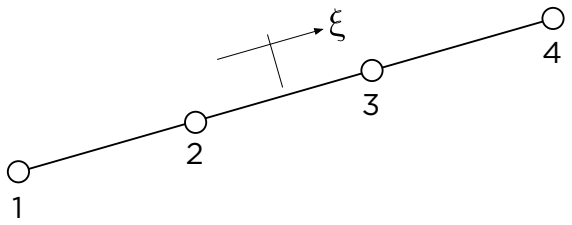


Figure Q2: 1D cubic finite element

In finite element analysis we need to determine the derivatives of the shape functions so that we can determine the strain within the elements and assemble the stiffness contribution of each element to the global system of equations.

(b) Without doing any additional derivations/calculations, for any point in the element shown in Figure Q2 what is

$$\sum_{i=1}^{4} \left(\frac{\partial N_i}{\partial \xi} \right)$$

equal to? Justify your answer with reference to the properties of shape functions. **[20%]**

(c) How many Gauss points should you use to integrate the stiffness of the element shown in Figure Q2? Justify your answer. **[15%]**

(d) The stiffness of a finite element can be expressed as

$$[k^e] = \int_V [B]^T [D^e][B] dV.$$

If the element shown in Figure Q2 was used to analyse a 3D problem, what would be the size and the units of each of the matrices in the above equation? **[15%]**

continued

Q3

Consider the finite element problem shown in Figure Q3 (a), where a unit square domain meshed by 4 equal-sized triangular elements is subjected to a traction on one edge. The imposed traction is equal to

$$\{t\} = \left\{ \begin{array}{c} t_x \\ t_y \end{array} \right\}, \qquad \text{where} \qquad t_x = \frac{1}{2} + y - \frac{6}{5}y^2 \text{ kN/m} \qquad \text{and} \qquad t_y = 0.$$

Three edges of the square are restrained by roller boundary conditions and node 5 is constrained so that it cannot move in the y direction. The shape functions for the three noded triangle shown in Figure Q3 (b) are

$$N_1 = 1 - \xi - \eta, \qquad N_2 = \eta \qquad \text{and} \qquad N_3 = \xi.$$

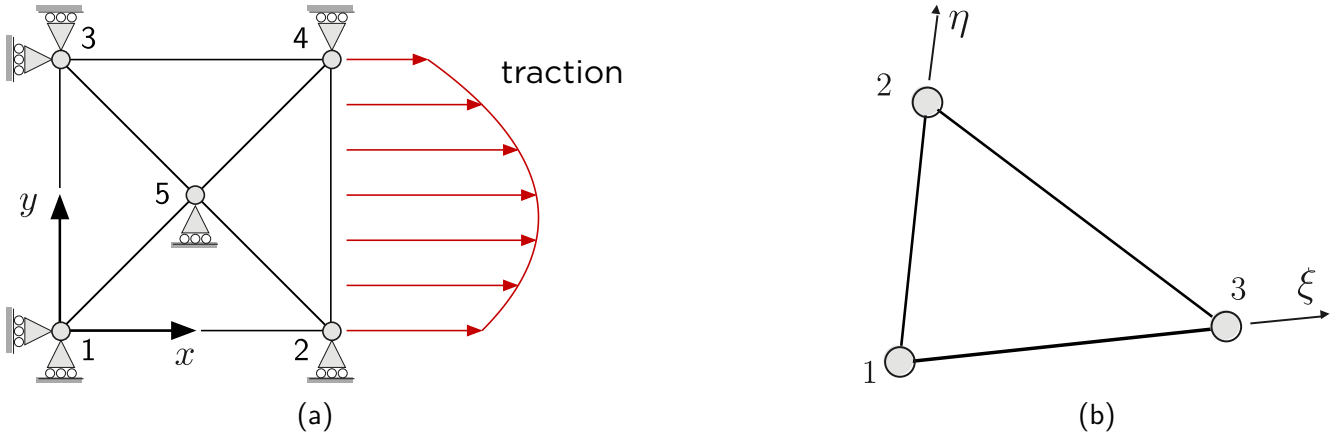


Figure Q3: (a) global mesh and (b) local node numbering where $(\xi, \eta) \in [0, 1]$

(a) Show that the equivalent nodal loads associated with the imposed traction at nodes 2 and 4 are

$$\{f_2\} = \left\{ \begin{array}{c} 0.3167 \\ 0 \end{array} \right\} \text{ kN} \qquad \text{and} \qquad \{f_4\} = \left\{ \begin{array}{c} 0.2833 \\ 0 \end{array} \right\} \text{ kN}$$

[40%]

(b) Given that the element topology matrix for the problem shown in Figure Q3 (a) is

$$\text{element topology} = \left[\begin{array}{ccc} 1 & 5 & 2 \\ 3 & 5 & 1 \\ 4 & 5 & 3 \\ 2 & 5 & 4 \end{array} \right],$$

the stiffness matrix, with the terms ordered according to the local convention shown in Figure Q3 (b), for elements 1 and 3 is

$$[k^e] = \left[\begin{array}{rrrrrr} 3.82 & 1.74 & -2.08 & -1.39 & -1.74 & -0.35 \\ 1.74 & 3.82 & -2.08 & -5.56 & 0.35 & 1.74 \\ -2.08 & -2.08 & 4.17 & 0.00 & -2.08 & 2.08 \\ -1.39 & -5.56 & 0.00 & 11.11 & 1.39 & -5.56 \\ -1.74 & 0.35 & -2.08 & 1.39 & 3.82 & -1.74 \\ -0.35 & 1.74 & 2.08 & -5.56 & -1.74 & 3.82 \end{array} \right] \text{ MN/m.}$$

continued

and the stiffness matrix for elements 2 and 4 is

$$[k^e] = \begin{bmatrix} 3.82 & -1.74 & -5.56 & 2.08 & 1.74 & -0.35 \\ -1.74 & 3.82 & 1.39 & -2.08 & 0.35 & -1.74 \\ -5.56 & 1.39 & 11.11 & 0.00 & -5.56 & -1.39 \\ 2.08 & -2.08 & 0.00 & 4.17 & -2.08 & -2.08 \\ 1.74 & 0.35 & -5.56 & -2.08 & 3.82 & 1.74 \\ -0.35 & -1.74 & -1.39 & -2.08 & 1.74 & 3.82 \end{bmatrix} \text{ MN/m.}$$

Determine the unknown displacements. **[50%]**

(c) How many Gauss points would you use to integrate the stiffness of each of the elements shown in Figure Q3 (a). Justify your answer. **[10%]**

ENGI3411-WE01 Solid Mechanics 3
Course: Fracture and Fatigue
Formula sheet

Stress and displacement distribution locally around the tip of a mode I crack oriented in the x-direction

$$\sigma_{xx} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 - \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{yy} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{xy} = \frac{K_I}{\sqrt{2\pi r}} \sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{3\theta}{2} + ...$$

$$u_x = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\cos\frac{\theta}{2}\left[\frac{1}{2}(\kappa - 1) + \sin^2\frac{\theta}{2}\right] + \cdots$$

$$u_y = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\sin\frac{\theta}{2}\left[\frac{1}{2}(\kappa + 1) - \cos^2\frac{\theta}{2}\right] + \cdots$$

$\kappa = 3 - 4\nu$ (plane strain), or $(3-\nu)/(1+\nu)$ (plane stress)

Mode I stress intensity factor

$$K_I = Y\sigma\sqrt{\pi a}$$

Internal crack of length $2a$ in infinite plate: $Y = 1$
Edge crack of length a in relatively large plate: $Y = 1.12$

Plastic zone correction

$$\text{Equivalent crack length} = a + \frac{K_I^2}{2\pi\sigma_Y^2}$$

Failure diagrams for mean stress correction

Modified Goodman:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_u}\right]$$

Gerber parabola:

$$S_a = S_D\left[1 - \left(\frac{S_{mean}}{S_u}\right)^2\right]$$

Soderberg:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_{yield}}\right]$$

Strain-life method

Strain life equation:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'}{E}(2N_f)^b + \varepsilon_f'(2N_f)^c$$

Mean stress correction (Smith Watson Topper):

$$\left(\sigma_{mean} + \frac{\Delta\sigma}{2}\right)\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'^2}{E}(2N_f)^{2b} + \sigma_f'\varepsilon_f'(2N_f)^{b+c}$$

continued

Macro crack growth laws based on fracture mechanics approach

Paris Law:

$$\frac{da}{dN} = C(\Delta K)^m$$

Broek and Schijve:

$$\frac{da}{dN} = CK_{max}^2 \Delta K$$

Walker:

$$\frac{da}{dN} = CK_{max}^m (\Delta K)^n$$

Walker:

$$\frac{da}{dN} = C\overline{\Delta K}^n \qquad \text{where } \overline{\Delta K} = S_{max}(1-R)^m\sqrt{\pi a}$$

Forman:

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)(K_{Ic} - K_{max})}$$

continued

ENGI3411-WE01 Solid Mechanics 3
Course: Finite-element method
Formula sheet

Three-dimensional stress-strain relationship

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & \nu & 0 & 0 & 0 \\ \nu & (1-\nu) & \nu & 0 & 0 & 0 \\ \nu & \nu & (1-\nu) & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix}$$

Two-dimensional stress-strain relationships

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & 0 \\ \nu & (1-\nu) & 0 \\ 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane strain)}$$

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{(1-\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane stress)}$$

Three-dimensional strain definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = [L]\{u\} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 \\ 0 & \frac{\partial}{\partial y} & 0 \\ 0 & 0 & \frac{\partial}{\partial z} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix}$$

Interpolation using shape functions

$$u = \sum_{i=1}^{n_{en}} N_i u_i = \{N\}^T \{u\}$$

Three-dimensional strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial x} & 0 & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial y} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial z} & \dots & 0 & 0 & \frac{\partial N_{n_{en}}}{\partial z} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial y} & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial z} & \frac{\partial N_{n_{en}}}{\partial y} \\ \frac{\partial N_1}{\partial z} & 0 & \frac{\partial N_1}{\partial x} & \dots & \frac{\partial N_{n_{en}}}{\partial z} & 0 & \frac{\partial N_{n_{en}}}{\partial x} \end{bmatrix}}_{[B]=[L][N]} \begin{Bmatrix} u_1 \\ v_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ v_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}$$

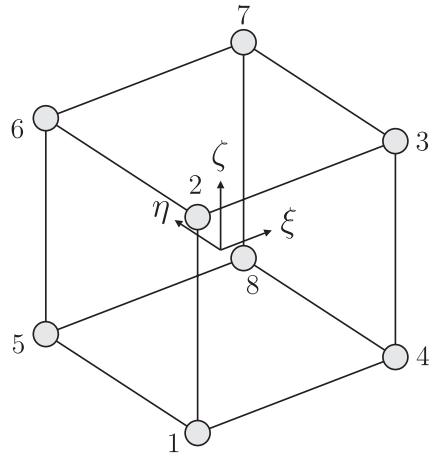
continued

Plane-stress/strain strain-displacement definition

$$\left\{\begin{array}{c}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \gamma_{xy}\end{array}\right\} = \left[\begin{array}{cccc}\frac{\partial N_1}{\partial x} & 0 & \cdots & \frac{\partial N_{n_{en}}}{\partial x} & 0\\ 0 & \frac{\partial N_1}{\partial y} & \cdots & 0 & \frac{\partial N_{n_{en}}}{\partial y}\\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \cdots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x}\end{array}\right] \left\{\begin{array}{c}u_1\\ v_1\\ \vdots\\ u_{n_{en}}\\ v_{n_{en}}\end{array}\right\}.$$

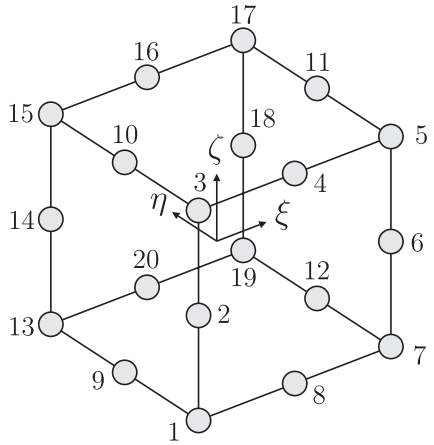
Shape functions for hexahedral elements

Tri-linear eight-noded hexahedral



$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)/8\\
N_2 &= (1-\xi)(1-\eta)(1+\zeta)/8\\
N_3 &= (1+\xi)(1-\eta)(1+\zeta)/8\\
N_4 &= (1+\xi)(1-\eta)(1-\zeta)/8\\
N_5 &= (1-\xi)(1+\eta)(1-\zeta)/8\\
N_6 &= (1-\xi)(1+\eta)(1+\zeta)/8\\
N_7 &= (1+\xi)(1+\eta)(1+\zeta)/8\\
N_8 &= (1+\xi)(1+\eta)(1-\zeta)/8
\end{aligned}$$

Tri-quadratic twenty-noded hexahedral



$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)(-\xi-\eta-\zeta-2)/8\\
N_2 &= (1-\xi)(1-\eta)(1-\zeta^2)/4\\
N_3 &= (1-\xi)(1-\eta)(1+\zeta)(-\xi-\eta+\zeta-2)/8\\
N_4 &= (1-\xi^2)(1-\eta)(1+\zeta)/4\\
N_5 &= (1+\xi)(1-\eta)(1+\zeta)(\xi-\eta+\zeta-2)/8\\
N_6 &= (1+\xi)(1-\eta)(1-\zeta^2)/4\\
N_7 &= (1+\xi)(1-\eta)(1-\zeta)(\xi-\eta-\zeta-2)/8\\
N_8 &= (1-\xi^2)(1-\eta)(1-\zeta)/4\\
N_9 &= (1-\xi)(1-\eta^2)(1-\zeta)/4\\
N_{10} &= (1-\xi)(1-\eta^2)(1+\zeta)/4\\
N_{11} &= (1+\xi)(1-\eta^2)(1+\zeta)/4\\
N_{12} &= (1+\xi)(1-\eta^2)(1-\zeta)/4\\
N_{13} &= (1-\xi)(1+\eta)(1-\zeta)(-\xi+\eta-\zeta-2)/8\\
N_{14} &= (1-\xi)(1+\eta)(1-\zeta^2)/4\\
N_{15} &= (1-\xi)(1+\eta)(1+\zeta)(-\xi+\eta+\zeta-2)/8\\
N_{16} &= (1-\xi^2)(1+\eta)(1+\zeta)/4\\
N_{17} &= (1+\xi)(1+\eta)(1+\zeta)(\xi+\eta+\zeta-2)/8\\
N_{18} &= (1+\xi)(1+\eta)(1-\zeta^2)/4\\
N_{19} &= (1+\xi)(1+\eta)(1-\zeta)(\xi+\eta-\zeta-2)/8\\
N_{20} &= (1-\xi^2)(1+\eta)(1-\zeta)/4
\end{aligned}$$

continued

Axi-symmetric strain-displacement definition

$$\left\{\begin{array}{c}\varepsilon_{rr}\\ \varepsilon_{zz}\\ \varepsilon_{\theta\theta}\\ \gamma_{rz}\end{array}\right\}=\left[\begin{array}{ccccc}\frac{\partial N_1}{\partial r} & 0 & \cdots & \frac{\partial N_{nen}}{\partial r} & 0\\ 0 & \frac{\partial N_1}{\partial z} & \cdots & 0 & \frac{\partial N_{nen}}{\partial z}\\ \frac{N_1}{r_1} & 0 & \cdots & \frac{N_{nen}}{r_{nen}} & 0\\ \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial r} & \cdots & \frac{\partial N_{nen}}{\partial z} & \frac{\partial N_{nen}}{\partial r}\end{array}\right]\left\{\begin{array}{c}u_1\\ w_1\\ \vdots\\ u_{nen}\\ w_{nen}\end{array}\right\}.$$

Global shape function derivatives

$$\left\{\begin{array}{c}\frac{\partial N_i}{\partial x}\\ \frac{\partial N_i}{\partial y}\\ \frac{\partial N_i}{\partial z}\end{array}\right\}=\left[\begin{array}{ccc}\frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} & \frac{\partial \zeta}{\partial x}\\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} & \frac{\partial \zeta}{\partial y}\\ \frac{\partial \xi}{\partial z} & \frac{\partial \eta}{\partial z} & \frac{\partial \zeta}{\partial z}\end{array}\right]\left\{\begin{array}{c}\frac{\partial N_i}{\partial \xi}\\ \frac{\partial N_i}{\partial \eta}\\ \frac{\partial N_i}{\partial \zeta}\end{array}\right\}.$$

The Jacobian matrix

$$[J]=\left[\begin{array}{ccc}\frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi}\\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta}\\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta}\end{array}\right]=\left[\begin{array}{cccc}\frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \cdots & \frac{\partial N_{nen}}{\partial \xi}\\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \cdots & \frac{\partial N_{nen}}{\partial \eta}\\ \frac{\partial N_1}{\partial \zeta} & \frac{\partial N_2}{\partial \zeta} & \cdots & \frac{\partial N_{nen}}{\partial \zeta}\end{array}\right]\left[\begin{array}{ccc}x_1 & y_1 & z_1\\ x_2 & y_2 & z_2\\ \vdots & \vdots & \vdots\\ x_{n_{en}} & y_{n_{en}} & z_{n_{en}}\end{array}\right]$$

Weak form of equilibrium

$$\int_V [B]^T[D^{\mathsf{e}}][B]dV\{d\}=\int_S [N]^T\{t\}dS+\int_V [N]^T\{f^b\}dV.$$

Element stiffness and force vector

$$[k^e]=\int_V [B]^T[D^{\mathsf{e}}][B]dV \qquad \text{and} \qquad \{f^e\}=\int_S [N]^T\{t\}dS+\int_V [N]^T\{f^b\}dV.$$

Gauss-Legendre quadrature

$$\int_{-1}^{1} f(x)dx \approx \sum_{i=1}^{n_{Gp}} f(x_i)w_i$$

n_{Gp}	Abscissae, x_i	Weights, w_i
1	0.000000000000000	2.000000000000000
2	±0.577350269189626	1.000000000000000
3	0.000000000000000	0.888888888888889
	±0.774596669241483	0.555555555555556
4	±0.339981043584856	0.652145154862546
	±0.861136311594053	0.347854845137454

Numerical approximation of the element stiffness matrix

$$[k^e] \approx \sum_{i=1}^{n_{Gp}} [B]_i^T[D^{\mathsf{e}}][B]_i w_i \det([J]_i)$$

END